

Initial Antenna Design Calculations

This document outlines the initial theoretical calculations for a dual-band reconfigurable microstrip patch antenna targeting the Wi-Fi bands at $f_1 = 2.45\text{ GHz}$ and $f_2 = 5.8\text{ GHz}$

1. Substrate Selection

Parameter	Value	Unit	Rationale
Substrate Material	Rogers RT/duroid 5880	-	Low loss tangent and low dielectric constant for high-frequency performance.
Dielectric Constant	2.2	-	Standard value for RT/duroid 5880.
Substrate Thickness	1.57	mm	Standard thickness
Loss Tangent	0.0009	-	Low loss.
Copper Thickness	0.035	mm	Standard 1 oz copper.

2. Target Frequencies

The antenna is designed to resonate at two frequencies:

- f_1 (Lower Band): 2.45 GHz (Wi-Fi 2.4 GHz)
- f_2 (Upper Band): 5.8 GHz (Wi-Fi 5 GHz)

The initial patch dimensions will be calculated based on the lower frequency, $f_1 = 2.45\text{ GHz}$

3. Patch Dimensions Calculation (for $f_1 = 2.45\text{ GHz}$)

3.1. Patch Width (W)

The width W is calculated to be λ_{f_1} , which ensures efficient radiation.

Where $c = 3 \times 10^8\text{ m/s}$ (speed of light).

$$\{2 \times 2.45 \times 10^9 \text{ Hz} \times \sqrt{\frac{2.2 + 1}{2}}\} \approx 48.78 \text{ mm}$$

3.2. Effective Dielectric Constant (ϵ_{re})

The effective dielectric constant accounts for the fringing fields.

$$\epsilon_{re} = \epsilon_r + 1 + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{W} \right]^{-0.5}$$

With $h/W = 1.57 / 48.78 \approx 0.032$:

$$\epsilon_{re} = \frac{2.2 + 1}{2} + \frac{2.2 - 1}{2} \left[1 + 12 (0.032) \right]^{-0.5} \\ \approx 1.6 + 0.6 \times 0.887 \approx 2.132$$

3.3. Effective Length (L_{eff})

The effective length is the length of the patch considering the effective dielectric constant.

$$L_{eff} = \frac{c}{2 f_r \sqrt{\epsilon_{re}}} = \frac{3 \times 10^8 \text{ m/s}}{2 \times 2.45 \times 10^9 \text{ Hz} \times \sqrt{2.132}} \approx 41.86 \text{ mm}$$

3.4. Length Extension (ΔL)

The length extension due to fringing fields is:

$$\Delta L = 0.412 h \frac{(\epsilon_{re} + 0.3)(W/h + 0.264)}{(\epsilon_{re} - 0.258)(W/h + 0.8)}$$

With $W/h \approx 31.07$:

$$\Delta L = 0.412 (1.57) \frac{(2.132 + 0.3)(31.07 + 0.264)}{(2.132 - 0.258)(31.07 + 0.8)} \\ \approx 0.646 \times \frac{2.432 \times 31.334}{1.874 \times 31.87} \approx 1.61 \text{ mm}$$

3.5. Patch Length (L)

The actual length of the patch is:

$$L = L_{eff} - 2 \Delta L = 41.86 \text{ mm} - 2 (1.61 \text{ mm}) \approx 38.64 \text{ mm}$$

4. Ground Size

A standard ground plane size ($L_g \times W_g$) is chosen to be at least $6h$ larger than the patch dimensions on all sides. For simulation, a square ground plane is often used.

$$L_g = W_g = 100 \text{ {mm}}$$

5. Feed Point/Slot Sizing

An inset-fed microstrip line is used for 50 Ω impedance matching.

5.1. Microstrip Feed Line Width (W_f)

For a 50 Ω line on RT/duroid 5880 ($\epsilon_r = 2.2$, $h = 1.57 \text{ {mm}}$), the characteristic impedance calculation yields:

$$W_f \approx 4.8 \text{ {mm}}$$

5.2. Inset Feed Length (L_{in})

The inset feed length L_{in} is the distance from the edge of the patch to the feed point, which determines the input impedance. For a 50 Ω match, the feed point is typically close to the edge. An initial estimate is based on the half-wavelength impedance variation:

Where $R_{in}(y=0)$ is the edge resistance, approximately $200 \text{ }\Omega$. We want $R_{in}(y=L_{in}) = 50 \text{ }\Omega$.

$$L_{in} = \frac{L}{\pi} \cos^{-1} \left(\sqrt{\frac{50}{200}} \right) = \frac{38.64 \text{ {mm}}}{\pi} \cos^{-1} (0.5) \approx 12.29 \text{ {mm}} \times 1.047 \text{ {rad}} \approx 12.87 \text{ {mm}}$$

5.3. Inset Gap Width (g)

The gap between the patch and the feed line is typically small, e.g., $g = 0.5 \text{ {mm}}$.

Summary of Initial Design Parameters

Parameter	Symbol	Value	Unit
Substrate Dielectric Constant	ϵ_r	2.2	-

Parameter	Symbol	Value	Unit
Substrate Thickness	h	1.57	mm
Target Frequency	f_r	2.45	GHz
Patch Width	W	48.78	mm
Patch Length	L	38.64	mm
Ground Plane Size	$L_g \times W_g$	100×100	mm
Feed Line Width	W_f	4.8	mm
Inset Feed Length	L_{in}	12.87	mm
Inset Gap Width	g	0.5	mm

This design will serve as the **Base Antenna** for the next simulation phase. The dual-band and reconfigurable nature will be introduced in Phase 3.